

JUNGUANG JIA

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EDUCATION

Columbia University <i>M.A. in Statistics, Theory & Methods Honors Track</i>	<i>Sep 2025 – Present</i>
University of California, San Diego <i>B.S. in Mathematics (Probability & Statistics)</i> Provost Honors (multiple quarters)	<i>Sep 2020 – Jul 2024</i>
Fudan University <i>B.S. in Statistics (Term Exchange)</i>	<i>Aug 2022 – Jun 2023</i>

ACADEMIC DISTINCTIONS

Theory & Methods Core Competency Examination *May 2026*

Passed the required examination for the Theory & Methods Honors Track, a rigorous theory-focused curriculum in probability and mathematical statistics with research experience for doctoral preparation.

RESEARCH & PROJECT EXPERIENCE

Stochastic Trace Estimation: Hutchinson and Hutch++ *Spring 2026 – Present*

Computational Statistics Project, Columbia University

Instructor: [Christopher R. Harshaw](#)

- Reconstructed unbiasedness and complexity arguments for Hutchinson and Hutch++; examined randomized low-rank sketching as a variance-reduction strategy.
- Implemented reproducible Julia experiments for structured positive semidefinite (PSD) matrices and graph triangle counting, using dense and implicit sparse matrix-vector operators, seeded Monte Carlo trials, and unit tests.
- On a $1,000 \times 1,000$ exponential-spectrum PSD matrix, Hutch++ reached a 1% mean relative-error target at 30 matrix-vector products, whereas Hutchinson did not reach it by 120 (1,000 Monte Carlo replications).

ArtVenn – Cultural Heritage Data & Community Platform *Oct 2024 – Present*

Founder & Technical Project Lead

- Conducted field documentation across 200+ cities in China, recording geospatial coordinates and analyzing surviving textual and symbolic traces to build structured location and condition data.
- Designed relational catalog schemas with provenance, geospatial, and source metadata; developed validated data-import and editorial workflows with transactional updates and controlled publication.
- Built the platform with TypeScript, Next.js/React, Node.js, and PostgreSQL; implemented OpenAPI contracts, Chinese text normalization, ranked search, and object-storage integration.
- Ongoing work focuses on organizing the collected corpus and preparing it for downstream statistical and machine-learning analysis.

SELECTED COURSEWORK

Ph.D.-Level Coursework – Columbia University

Computational Statistics (STAT GR6104)

Theoretical Statistics I (STAT GR6201)

In progress

Probabilistic Models and Machine Learning (STCS GR6701)

In progress

Optimization I (IEOR E6613)

In progress

Other Selected Coursework

Honors Probability Theory

Honors Statistical Inference

Honors Linear Regression Models

Bayesian Statistics

Stochastic Processes and Applications I

Time Series Analysis

Introduction to Analysis II

Unsupervised Learning (in progress)

Theory of Diffusion Models (in progress)

Advanced Data Analysis (in progress)

WORK EXPERIENCE

Mobalytics

Jul 2023 – Sep 2023

Data Analyst, Data Department

California, USA

- Analyzed demographic and player-behavior data across three games using SQL, correlation, and regression analysis to examine engagement and personalization opportunities.
- Built Tableau visualizations and player-segmentation analyses; translated findings into recommendations for the marketing team.

Shanghai Dingbang International Trading Co., Ltd.

Sep 2022 – Dec 2022

Financial Assistant, Production Statistics Department

Shanghai, China

- Maintained production-statistics records and compiled daily reports to identify deviations; coordinated reporting schedules to reduce end-of-period workloads.

Hua An Securities

Jul 2022 – Sep 2022

Investment Research Analyst, Investment Research Department

Shanghai, China

- Conducted company and industry research in the catalyst sector; applied multiple linear regression and principal component analysis to interpret market dynamics and support investment recommendations.

Shanghai Yunyi Huan Yu Technology Co., Ltd.

Jun 2021 – Aug 2021

R&D Assistant, Data Analysis Department

Shanghai, China

- Supported Python/Spark data integration, model analysis, and validation for O2O and SaaS platforms serving small and medium-sized enterprises and e-commerce businesses.

TECHNICAL SKILLS

Programming Python, R, Julia, SQL, TypeScript

Data & Web PostgreSQL, Apache Spark, Tableau, Next.js, React, Node.js

Tools Git, Docker, Linux, GitHub Actions, L^AT_EX